



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B.Sc Chemistry				
Course Title	ADVANCED TOPICS IN INORGANIC CHEMISTRY				
Type of Course	ELECTIVE IN MAJOR				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	IR spectroscopy Types of ligands Preliminary idea about ESR, Mossbauer and NMR spectra of complexes Ligand exchange reactions Types of magnetic properties of complexes				
Course Summary	This course gives an insight to the application of IR spectroscopy in coordination complexes This course helps to analyse complexes using ESR, Mossbauer and NMR spectra It provides the knowledge of fascinating applications of complexes in medical field This course explains the anomalous magnetic properties of complexes This course enables the student to analyse different characteristics of a complex using various spectroscopic techniques				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand and apply infrared spectroscopy techniques for characterizing coordination compounds	Ap	C	Instructor-created exams / Assignments
CO2	Interpret the role of ESR and Mossbauer spectroscopic techniques in	Ap	C	Assignment / seminar/Quiz

	elucidating complex structures			
CO3	Evaluate the application of metal complexes in the medicinal field and their interaction with biological entities	E	C	Assignment/Seminar/Class test
CO4	Analyze the significance of anomalous magnetic moments in understanding complex structures and dynamics	An	C	Class Test/ Assignment/Viva Voce
CO5	Perform spectral analysis of complexes and solve related problems using IR, ESR and Mossbauer techniques	Ap	P	Group work /Assignment/class test/
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Mark
I	CHARACTERIZATION OF COORDINATION COMPLEXES USING IR SPECTROSCOPIC TECHNIQUES		13	26
	1	Infrared spectra of metal complexes. Group frequency concept.	1	
	2	Changes in ligand vibrations on coordination-	2	
	3	Effect of complex formation on symmetry, electronic structure and bonding characteristics of ligands –	4	
	4	Coordination of nitrate ion, sulphate ion, acetate ion, perchlorate ions, cyano group - metal ligand vibrations – water as a ligand –	3	
	5	IR spectroscopy of carbonyl complexes -	1	

	6	Application of IR spectroscopy in coordination complexes –	2	
II	CHARACTERIZATION OF COORDINATION COMPLEXES USING ESR AND MOSSBAUER SPECTROSCOPIC TECHNIQUES		10	22
	1	ESR spectra- Introduction	1	
	2	Importance of g values in structure elucidation of complexes	1	
	3	Application of ESR measurements to magnetically dilute and concentrated complexes -	2	
	4	Mossbauer spectra - application to iron complexes –	2	
	5	Factors affecting the chemical shift in coordination complexes	2	
	6	NMR spectra of diamagnetic copper complexes	2	
III	APPLICATIONS OF METAL COMPLEXES IN MEDICINAL FIELD		15	28
	1	Introduction – DNA-metal complex interaction –	2	
	2	Effect of Ligand exchange reactions and redox reactions in biological activity of metal complexes -	2	
	3	Effect of catalytic activity and photo physical activity on biological activity	2	
	4	Virtual screening of pharmacological behaviour - Drug likeness and bioavailability	2	
	5	Lipinski's Rule of 5 - Pharmacokinetic analysis of a drug molecule –ADMET	2	
	6	Analysis – In vitro and in vivo studies	2	
	7	Molecular docking.	1	
	8	Biological activities of transition metal complexes of Schiff bases, aromatic hydrazones	2	
IV	ANOMALOUS MAGNETIC MOMENTS OF METAL COMPLEXES		10	22
	1	Introduction – Equilibrium between two spin states	1	
	2	Magnetically non-equivalent sites in the metal ions – solute-solvent interactions	1	
	3	solute-solute interaction – configurational equilibrium – Antiferromagnetism – types –	2	
	4	antiferromagnetic exchange pathways –examples of antiferromagnetic binuclear complexes (Cu(II), V(IV))	2	

	5	Binuclear complexes with non-equivalent ions	2	
	6	Ferromagnetism – Trinuclear complexes	2	
V	SPECTRAL ANALYSIS OF COMPLEXES (Open ended)		12	
	IR, ESR and Mossbauer spectral analysis of some complexes (problem solving)			

REFERENCES

1. Concise Coordination Chemistry, 1/e, R Gopalan & V Ramalingam, Vikas Publishing
2. C.N. Banwell, Fundamentals of molecular Spectroscopy, 3rd ed. TMH, New Delhi, 1983.
3. . E.A.V. Ebsworth, Structural Methods in Inorganic Chemistry, 3rd ed., ELBS, Great Britain, 1987.
4. Drago, R. S. Physical Methods in Chemistry W. B. Saunders: Philadelphia, 1977.
5. Elements of Magnetochemistry by R L Dutta and A Shyamal, Edition, 2 ; Publisher, Affiliated East-West Press, 1993 ; ISBN, 818533692X, 9788185336923
6. Textbook of Drug Design and Discovery, Edited by Kristian Strømgaard Povl Krogsgaard-Larsen Ulf Madsen, CRC Press Taylor & Francis Group.
7. A Closer Look at Coordination Complexes, Sandeep Kaur-Ghumaan, Series: Chemistry Research and Applications, BISAC: SCI013030, DOI: <https://doi.org/10.52305/ENZL4915>

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
C O 1	3	1	2				3		1		1		1
C O 2	3	1		1	2		3		1		1		2
C O 3	3				2	2	3		1		2		3
C O 4						2	3		1		2		3
C O 5						2	3		1		3		3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Pr actical Exam	Assignmen t /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓

MINOR COURSES